

Supplementary Material 3: Notation

Notation	Description
n, p, k	number of observations, variables, embedding dimension, respectively
$\mathbf{X}$	$n \times p$ -dimensional data
$\mathbf{x}_{i*}$	row i of $\mathbf{X}$
$\mathbf{x}_{*j}$	column j of $\mathbf{X}$
$\mathbf{x}^*$	new p -dimensional observation ($1 \times p$)
$\widehat{\mathbf{X}}$	$n \times p$ -dimensional fitted values
$\mathbf{Y}$	$n \times k$ -dimensional data (layout)
$\mathbf{y}_{i*}$	row i of $\mathbf{Y}$
$\mathbf{y}_{*j}$	column j of $\mathbf{Y}$
$\widehat{\mathbf{Y}}$	$n \times k$ -dimensional fitted values
$\mathbf{y}^*$	new k -dimensional observation ($1 \times k$)
e_i	Residual of observation i , computed on p -dimensional data
P	orthonormal basis, generating a d -dimensional linear projection of p -dimensional data
T	true model
g	functional mapping from p -D to k -D, especially as prescribed by NLDR
θ	(Hyper-) parameters for NLDR method
r	ranges of the embedding components
$\mathbf{c}_{h*}^{(2)}$	coordinates of bin centers in 2-D
$\mathbf{c}_{h*}^{(p)}$	coordinates of bin centers in p -D
(b_1, b_2)	number of bins in each direction
(a_1, a_2)	binwidths, distance between centroids in each direction
(s_1, s_2)	starting coordinates of the hexagonal grid
q	buffer to ensure hexgrid covers data, proportion of data range, 0-1
m	number of non-empty bins
b	number of hexagons in the grid
h	hexagonal id
u_i	unique id for bin i , $i = 1, \dots, h$
H_h	set of observations in bin h
l	side length
A	area
n_h	number of points in hexagon h (bin count)
w_h	standardized number of points in hexagon h (standardized bin counts)
g	mapping from p to k dimensions
g^-	pseudo-inverse of g
$\hat{g}$	estimate of g

Table 1: Summary of notation for describing new methodology.